DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO. 5

Case study: Airline Management System

Mini Project: Centralized Club Management System

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Case Study: Airline Management System

Aim: Write cursor code using control structures and exception handling.

A cursor in SQL is a database object used to retrieve and process data row-by-row from a result set.

Normally SQL works on multiple rows at once, but a cursor allows:

- Sequential (one-by-one) processing
- Applying complex logic on each row

Why use Cursor?

- When operations depend on previous row values
- When row-by-row calculations are required
- When complex conditions cannot be handled by simple queries

Control Structures Used

Control structures are used to control flow of execution:

- **LOOP / WHILE** → repeat execution
- **IF / ELSEIF / ELSE** → decision making
- **LEAVE** → exit loop

Exception Handling

In MySQL, exception handling is done using:

DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;

It handles the condition when no more rows are available in cursor.

Steps to Use Cursor

1. Declare variables
2. Declare cursor
3. Declare handler
4. Open cursor
5. Fetch rows
6. Apply logic (IF/LOOP)
7. Close cursor

```
mysql> select * from airline;

+------------+--------------+---------+--------------
+

| airline_id | airline_name | country |
founded_year |

+------------+--------------+---------+--------------
+

|          1 | Air India    | India   |         NULL |

|          2 | Indigo       | India   |         NULL |

|          3 | Spicejet     | India   |         NULL |

|          4 | Airindia     | India   |         NULL |

+------------+--------------+---------+--------------
+

4 rows in set (0.03 sec)


mysql> select * from booking;

+------------+-----------+--------------+-------------
-+---------+--------------+

| booking_id | flight_id | passenger_id |
booking_date | seat_no | travel_class |

+------------+-----------+--------------+-------------
-+---------+--------------+

|       1001 |       101 |          501 | 2024-02-
01   | 12A     | Economy      |

|       1002 |       105 |          789 | 2023-09-
08   | 14B     | Economy      |

|       1004 |       101 |          678 | 2025-01-
01   | 15A     | Economy      |

|       5003 |       101 |          501 | 2025-05-
01   | 10A     | Business     |

+------------+-----------+--------------+-------------
-+---------+--------------+
```

```
4 rows in set (0.04 sec)


mysql> select * from flight;

+-----------+------------+---------+-------------+---
-------------+--------------+-------------+----------
--+-----------+

| flight_id | airline_id | source  |
destination | departure_time |
arrival_time | flight_type | total_seats |
status    |

+-----------+------------+---------+-------------+---
-------------+--------------+-------------+----------
--+-----------+

|       101 |          1 | Mumbai  | Lucknow
| 10:00:00       | 12:00:00     | Domestic    |
180 | Delayed   |

|       102 |          2 | kolkata | nagpur      |
12:00:00       | 04:00:00     | Domestic    |
180 | Scheduled |

|       104 |          4 | mumbai  | chennai     |
06:00:00       | 09:00:00     | Domestic    |
180 | Scheduled |

+-----------+------------+---------+-------------+---
-------------+--------------+-------------+----------
--+-----------+

3 rows in set (0.02 sec)


mysql> select * from passenger;

+--------------+----------------+--------+------+----
---------+-------------+

| passenger_id | passenger_name |
gender | age  | passport_no | nationality
|
```

```
+--------------+----------------+--------+------+----
---------+-------------+

|        501 | Rahul Mehta    | M      |   35 |
P1234567    | Indian      |

|        678 | Rohit          | M      |   23 |
P85666      | Indian      |
```

```
|        888 | Jiya           | F      |   46 |
P45677      | Indian      |

+--------------+----------------+--------+------+----
---------+-------------+

3 rows in set (0.01 sec)
```

1. Cursor: Upgrade Ticket + Service Category

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE UpgradeBooking()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE b_id INT;
    ->     DECLARE price INT;
    ->     DECLARE t_class VARCHAR(20);
    ->     DECLARE service VARCHAR(30);
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT booking_id, ticket_price, travel_class FROM booking;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO b_id, price, t_class;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF t_class = 'Business' THEN
    ->             SET price = price + 5000;
    ->             SET service = 'Business';
    ->         ELSEIF t_class = 'Economy' THEN
    ->             SET service = 'Basic';
    ->         END IF;
    ->
    ->         UPDATE booking
    ->         SET ticket_price = price,
    ->             service_category = service
    ->         WHERE booking_id = b_id;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    -> END //
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CALL UpgradeBooking();
Query OK, 0 rows affected (0.07 sec)

mysql> SELECT booking_id, travel_class, ticket_price, service_category
    -> FROM booking;
+------------+--------------+--------------+------------------+
| booking_id | travel_class | ticket_price | service_category |
+------------+--------------+--------------+------------------+
|       1001 | Economy      |        10000 | Basic            |
|       1002 | Economy      |        10000 | Basic            |
|       1004 | Economy      |        10000 | Basic            |
|       5003 | Business     |        15000 | Business         |
+------------+--------------+--------------+------------------+
4 rows in set (0.00 sec)
```

2. Cursor: Count Economy Bookings

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CountEconomy()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE t_class VARCHAR(20);
    ->     DECLARE total INT DEFAULT 0;
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT travel_class FROM booking;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO t_class;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF t_class = 'Economy' THEN
    ->             SET total = total + 1;
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    ->     SELECT total AS Total_Economy;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL CountEconomy();
+---------------+
| Total_Economy |
+---------------+
|             3 |
+---------------+
1 row in set (0.00 sec)
```

3. Cursor: Calculate Surcharge + Log

```
mysql> CREATE TABLE surcharge_log (
    ->     booking_id INT,
    ->     surcharge INT,
    ->     total_surcharge INT
    -> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CalcSurcharge()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE b_id INT;
    ->     DECLARE price INT;
    ->     DECLARE surcharge INT;
    ->     DECLARE total INT DEFAULT 0;
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT booking_id, ticket_price FROM booking;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO b_id, price;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF price > 12000 THEN
    ->             SET surcharge = price * 0.10;
    ->             SET total = total + surcharge;
    ->
    ->             INSERT INTO surcharge_log
    ->             VALUES (b_id, surcharge, total);
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL CalcSurcharge();
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> CALL CalcSurcharge();
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM surcharge_log;
+------------+-----------+-----------------+
| booking_id | surcharge | total_surcharge |
+------------+-----------+-----------------+
|       5003 |      1500 |            1500 |
+------------+-----------+-----------------+
1 row in set (0.00 sec)
```

Mini Project : Centralized Club Management System

```
mysql> select * from attendance;

+---------------+-----------+----------+---------+

| attendance_id | member_id | event_id | status  |

+---------------+-----------+----------+---------+

|             2 |         2 |        1 | Absent  |

|             3 |         3 |        2 | Present |

|             4 |         4 |        3 | Present |

|             5 |         5 |        4 | Absent  |

+---------------+-----------+----------+---------+

4 rows in set (0.02 sec)


mysql> select * from budget_request;

+-----------+----------+---------------+---------+--
-------+------------+-----------------+

| budget_id | event_id | coordinator_id | club_id | lead_id | submission | approved_status |

+-----------+----------+---------------+---------+--
-------+------------+-----------------+

|         1 |        1 |           101 |       1 |     201 | 2026-04-01 | Pending        |

|         2 |        2 |           102 |       2 |     202 | 2026-04-05 | Approved       |

|         3 |        3 |           103 |       3 |     203 | 2026-04-07 | Pending        |

|         4 |        4 |           104 |       4 |     204 | 2026-04-10 | Approved       |

|         5 |        5 |           105 |       5 |     205 | 2026-04-12 | Pending        |

+-----------+----------+---------------+---------+--
-------+------------+-----------------+

5 rows in set (0.02 sec)


mysql> select * from club;

+---------+---------------+---------+----------------
+

| club_id | club_name     | lead_id | coordinator_id |

+---------+---------------+---------+----------------
+

|       1 | Tech Club     |     201 |            101 |

|       2 | Cultural Club |     202 |            102 |
```

```
|      3 | Sports Club   |     203 |          103
|

|      4 | Music Club    |     204 |          104
|

|      5 | Art Club      |     205 |          105 |

+---------+---------------+---------+----------------
+

5 rows in set (0.01 sec)


mysql> select * from club_coordinator;

+----------------+------------------+-----------------
+----------+---------+

| coordinator_id | coordinator_name |
email          | password | club_id |

+----------------+------------------+-----------------
+----------+---------+

|          101 | Anita           |
anita@gmail.com | pass     |      1 |

|          102 | Priya           |
priya@gmail.com | pass     |      2 |

|          103 | Rohit           |
rohit@gmail.com | pass     |      3 |

|          104 | Sneha           |
sneha@gmail.com | pass     |      4 |

|          105 | Vikas           |
vikas@gmail.com | pass     |      5 |

+----------------+------------------+-----------------
+----------+---------+

5 rows in set (0.02 sec)


mysql> select * from club_lead;

+---------+-----------+------------------+----------
+---------+----------------+

| lead_id | lead_name | email           |
password | club_id | coordinator_id |

+---------+-----------+------------------+----------
+---------+----------------+

|     201 | Rahul     | rahul@gmail.com  |
pass     |      1 |          101 |

|     202 | Karan     | karan@gmail.com  |
pass     |      2 |          102 |

|     203 | Amit      | amit@gmail.com   |
pass     |      3 |          103 |

|     204 | Pooja     | pooja@gmail.com  |
pass     |      4 |          104 |

|     205 | Neeraj    | neeraj@gmail.com |
pass     |      5 |          105 |

+---------+-----------+------------------+----------
+---------+----------------+

5 rows in set (0.02 sec)


mysql> select * from club_members;

+------+--------+------------------+------------+----
---------+---------+---------+

| m_id | m_name | m_email         |
m_password | event_roll  | domain  |
club_id |

+------+--------+------------------+------------+----
---------+---------+---------+

|    1 | Aman   | aman@gmail.com   | 123
| Participant | AI      |      1 |

|    2 | Riya   | riya@gmail.com   | 123
| Volunteer   | Design  |      1 |
```

```
|    3 | Neha    | neha@gmail.com    | 123
| Participant | Dance    |       2 |

|    4 | Arjun   | arjun@gmail.com   | 123
| Player      | Cricket |       3 |

|    5 | Meera   | meera@gmail.com   | 123
| Singer      | Vocal    |       4 |

|    7 | Simran | simran@gmail.com | 123
| Participant | AI       |       1 |

+------+--------+------------------+------------+----
---------+---------+---------+

6 rows in set (0.02 sec)


mysql> select * from event;

+----------+----------------+------------+------------
+-----------+---------+---------+-----------+

| event_id | event_name     | start_date |
end_date   | report_id | lead_id | club_id
| budget_id |

+----------+----------------+------------+------------
+-----------+---------+---------+-----------+

|        1 | Hackathon      | 2026-04-01 |
2026-04-02 |      NULL |     201 |       1 |
NULL |

|        2 | Dance Fest     | 2026-04-05 |
2026-04-06 |      NULL |     202 |       2 |
NULL |

|        3 | Football Match | 2026-04-07 |
2026-04-08 |      NULL |     203 |       3 |
NULL |

|        4 | Music Night    | 2026-04-10 |
2026-04-11 |      NULL |     204 |       4 |
NULL |

|        5 | Art Exhibition | 2026-04-12 |
2026-04-13 |      NULL |     205 |       5 |
NULL |

+----------+----------------+------------+------------
+-----------+---------+---------+-----------+

5 rows in set (0.01 sec)


mysql> select * from reports;

+-----------+----------+---------+----------------+

| report_id | event_id | lead_id |
coordinator_id |

+-----------+----------+---------+----------------+

|        1 |        1 |     201 |            101 |

|        2 |        2 |     202 |            102 |

|        3 |        3 |     203 |            103 |

|        4 |        4 |     204 |            104 |

|        5 |        5 |     205 |            105 |

+-----------+----------+---------+----------------+

5 rows in set (0.01 sec)


mysql> select * from super_admin;

+----------+------------+------------------+----------
+----------------+

| admin_id | admin_name | email            |
password | overall_budget |

+----------+------------+------------------+----------
+----------------+

|        1 | Admin1     | admin1@gmail.com
| pass123  |      100000.00 |

|        2 | Admin2     | admin2@gmail.com
| pass123  |      150000.00 |
```

```
|      3 | Admin3    | admin3@gmail.com | pass123  |     200000.00 |

|      4 | Admin4    | admin4@gmail.com | pass123  |     120000.00 |

|      5 | Admin5    | admin5@gmail.com | pass123  |     180000.00 |

+----------+------------+-----------------+----------+---------------+
```

1. Cursor: Count Total Present Members

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CountPresent()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE status_val VARCHAR(20);
    ->     DECLARE total INT DEFAULT 0;
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT status FROM attendance;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO status_val;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF status_val = 'Present' THEN
    ->             SET total = total + 1;
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    ->     SELECT total AS Total_Present;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL CountPresent();
+---------------+
| Total_Present |
+---------------+
|             2 |
+---------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

2. Cursor: List All Club Members Name

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE ListMembers()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE name VARCHAR(50);
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT m_name FROM club_members;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO name;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         SELECT name AS Member_Name;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CALL ListMembers();
+-------------+
| Member_Name |
+-------------+
| Aman        |
+-------------+
1 row in set (0.00 sec)

+-------------+
| Member_Name |
+-------------+
| Riya        |
+-------------+
1 row in set (0.01 sec)

+-------------+
| Member_Name |
+-------------+
| Neha        |
+-------------+
1 row in set (0.01 sec)

+-------------+
| Member_Name |
+-------------+
| Arjun       |
+-------------+
1 row in set (0.02 sec)

+-------------+
| Member_Name |
+-------------+
| Meera       |
+-------------+
1 row in set (0.03 sec)

+-------------+
| Member_Name |
+-------------+
| Simran      |
+-------------+
1 row in set (0.04 sec)

Query OK, 0 rows affected (0.04 sec)
```

3. Cursor: Count Approved Budget Requests

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CountApprovedBudget()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE status_val VARCHAR(20);
    ->     DECLARE total INT DEFAULT 0;
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT approved_status FROM budget_request;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO status_val;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         IF status_val = 'Approved' THEN
    ->             SET total = total + 1;
    ->         END IF;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    ->
    ->     SELECT total AS Approved_Count;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL CountApprovedBudget();
+----------------+
| Approved_Count |
+----------------+
|              2 |
+----------------+
1 row in set (0.00 sec)
```

4. Cursor: Display Events with Club Name

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE EventWithClub()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE e_name VARCHAR(50);
    ->     DECLARE c_id INT;
    ->     DECLARE c_name VARCHAR(50);
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT event_name, club_id FROM event;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO e_name, c_id;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         SELECT club_name INTO c_name
    ->         FROM club
    ->         WHERE club_id = c_id;
    ->
    ->         SELECT CONCAT(e_name, ' - ', c_name) AS Info;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CALL EventWithClub();
+-----------------------+
| Info                  |
+-----------------------+
| Hackathon - Tech Club |
+-----------------------+
1 row in set (0.02 sec)

+---------------------------+
| Info                      |
+---------------------------+
| Dance Fest - Cultural Club |
+---------------------------+
1 row in set (0.02 sec)

+----------------------------+
| Info                       |
+----------------------------+
| Football Match - Sports Club |
+----------------------------+
1 row in set (0.03 sec)

+-------------------------+
| Info                    |
+-------------------------+
| Music Night - Music Club |
+-------------------------+
1 row in set (0.04 sec)

+--------------------------+
| Info                     |
+--------------------------+
| Art Exhibition - Art Club |
+--------------------------+
1 row in set (0.04 sec)

Query OK, 0 rows affected (0.05 sec)
```

5. Cursor: Count Members per Club

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CountMembersPerClub()
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE cid INT;
    ->     DECLARE total INT;
    ->
    ->     DECLARE cur CURSOR FOR
    ->         SELECT DISTINCT club_id FROM club_members;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     OPEN cur;
    ->
    ->     loop1: LOOP
    ->         FETCH cur INTO cid;
    ->
    ->         IF done = 1 THEN
    ->             LEAVE loop1;
    ->         END IF;
    ->
    ->         SELECT COUNT(*) INTO total
    ->         FROM club_members
    ->         WHERE club_id = cid;
    ->
    ->         SELECT CONCAT('Club ID: ', cid, ' Members: ', total) AS Info;
    ->
    ->     END LOOP;
    ->
    ->     CLOSE cur;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CALL CountMembersPerClub();
+----------------------+
| Info                 |
+----------------------+
| Club ID: 1 Members: 3 |
+----------------------+
1 row in set (0.03 sec)

+----------------------+
| Info                 |
+----------------------+
| Club ID: 2 Members: 1 |
+----------------------+
1 row in set (0.04 sec)

+----------------------+
| Info                 |
+----------------------+
| Club ID: 3 Members: 1 |
+----------------------+
1 row in set (0.05 sec)

+----------------------+
| Info                 |
+----------------------+
| Club ID: 4 Members: 1 |
+----------------------+
1 row in set (0.06 sec)

Query OK, 0 rows affected (0.07 sec)
```

Conclusion

The cursor implementation in both the Airline Management System and Club Management System demonstrates how row-by-row data processing can be effectively performed in SQL. Cursors are useful when operations require sequential handling of records, such as counting specific entries, updating values based on conditions, and performing calculations.

By using control structures like loops and conditional statements, along with exception handling using handlers, complex logic can be executed efficiently. In the airline system, cursors help manage booking-related operations, while in the club system, they assist in handling attendance, member data, and budget processing.

Although cursors are slower compared to set-based operations, they are essential in scenarios where step-by-step processing is required. Thus, this assignment highlights the importance of cursors in implementing advanced database logic and real-world applications.